

Advanced Risk Modeling, Claims & Coverage Extensions: Bajaj Allianz Motor Insurance

This research document analyzes the technical and actuarial logic governing Bajaj Allianz's motor insurance portfolio, with a focus on telematics-driven pricing and advanced claim settlement protocols.

1. Advanced Risk Modeling & Telematics

Bajaj Allianz has transitioned from static pricing to behavioral underwriting through its "DriveSmart" and "Pay As You Consume" (PAYC) models.

A. DriveSmart Telematics Logic

- **Data Collection:** Uses an OBD-II (On-Board Diagnostics) device to track real-time data:
 - **Driving Score:** Calculated based on hard braking, sudden acceleration, and sharp cornering.
 - **Engine Health:** Monitors diagnostic codes to predict potential mechanical failures.
- **Incentive Logic:** High "Driving Scores" translate directly into renewal discounts or "reward points" for car accessories, effectively creating a personalized risk-premium loop.

B. "Pay As You Drive" (PAYC) Logic

- **Usage-Based Pricing:** Policyholders select a kilometer slab (e.g., 3,000, 5,000, or 10,000 km).
- **Verification:** Activation requires an odometer reading via video/photo on the "Caringly Yours" app.
- **Exhaustion Protocol:** If the limit is reached, Own Damage (OD) cover is suspended. However, users can "top up" kilometers via an endorsement.
- **Third-Party Safety:** Importantly, Third-Party Liability remains active even if the kilometer limit is exhausted, maintaining legal compliance.

2. Advanced Claim Logic: Total Loss & OTS

Claim settlement in motor insurance follows specific mathematical thresholds to determine repair viability versus total compensation.

A. Constructive Total Loss (CTL)

- **Threshold Logic:** If the aggregate cost of retrieval and repair exceeds **75% of the IDV (Insured Declared Value)**, the vehicle is declared a CTL.

- **Payout Calculation:** The insurer pays the full IDV minus the compulsory deductible.
- **Return to Invoice (RTI) Extension:** If this rider is active, the insurer ignores the IDV and pays the **original invoice value** (including registration and road tax), bridging the gap created by annual depreciation.

B. Motor On-The-Spot (OTS) Settlement

- **Self-Survey Logic:** For minor accidental damages (typically up to ₹20,000), policyholders can bypass the physical surveyor.
- **AI Validation:** The user uploads photos of the damage; the AI-driven engine estimates repair costs and offers an immediate cash settlement directly to the user's bank account.

3. Depreciation & Add-on Mechanics

The "Standard Policy" logic assumes a vehicle is a depreciating asset, but add-ons can override this math.

A. Zero Depreciation (Depreciation Shield)

- **Standard Logic:** Insurers deduct 50% for plastic/rubber parts and up to 30% for fiber-glass during a claim.
- **Zero-Dep Logic:** This add-on waives these deductions. However, most Bajaj Allianz policies limit this to **two claims per year** to prevent moral hazard (negligent driving).
- **Age Cap:** Usually available only for vehicles up to 5 years old.

B. Specialized Add-on Triggers

- **Engine Protector:** Covers "Consequential Damage." Standard logic excludes engine repair if the driver tries to start a car submerged in water (hydrostatic lock). This rider overrides that exclusion, covering repair/replacement of internal child parts (pistons, connecting rods).
- **Consumables Cover:** Covers items "consumed" during repair (engine oil, coolants, nuts, bolts) which are otherwise excluded from standard indemnity.

4. Administrative & Compliance Logic

- **Section 157 (GR 17):** In case of a vehicle sale, the Third-Party insurance is deemed transferred automatically for 14 days, but the Own Damage (OD) cover requires a formal endorsement and transfer of the No Claim Bonus (NCB).
- **No Claim Bonus (NCB) Recovery:** Bajaj Allianz allows users to transfer NCB from an old vehicle to a new one, even if the new vehicle is from a different manufacturer, provided the policyholder remains the same.

Summary of Logic Parameters

Feature	Logic Parameter	Operational Effect
CTL Declaration	> 75% of IDV	Vehicle treated as scrap; full IDV paid.
OTS Payout	< ₹20,000	Instant approval via mobile app self-survey.
PAYC Limit	Odometer Tracking	Discounted premium for low-usage vehicles.
Hydrostatic Lock	Consequential Loss	Claim denied unless Engine Protector is active.